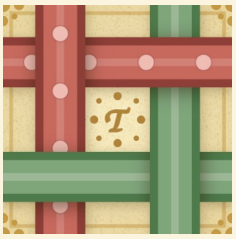
Repeating Knot Cards

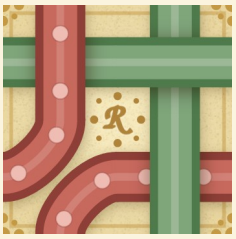
Justin Roughley, 2026

Four cards, nine wallpaper groups

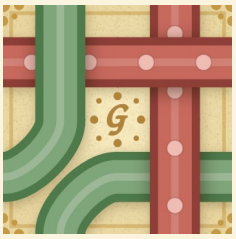
A pack of 60: forty-eight woven cards, nine target cards and three reference cards. Every card is a little patch of Celtic knotwork, and every one can be turned any way: wherever it goes, the ribbons still pass alternately over and under. Lay four cards in a square, repeat the square, and the knot takes on one of nine kinds of symmetry: nine of the seventeen wallpaper groups (en.wikipedia.org/wiki/Wallpaper_group).



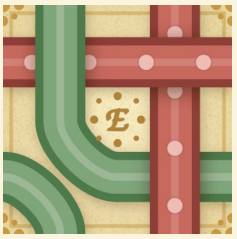
T × 16
four crossings: a trellis



R × 8
two red bends



G × 16
two green bends



E × 8
an elbow and a corner

HOW TO PLAY

1. Lay four cards in a square, each turned any way you like.
2. Make three more squares the same and join them: sixteen cards. Colours must match across every edge.
3. Look for mirrors, glides and turns, and name the pattern's wallpaper group. Over and under count; colours do not.

THREE CHALLENGES

EASY

Study a group

Take a target card and read its back first: it names the group and marks its mirrors, glides and turns on one square of four. Build the pattern from the front and find each mark in it.

MEDIUM

Name the group

Take a target card, build the pattern from the four letters on its front, each turned as shown, and work out its group. Turn the card over to check.

HARD

Find all nine

With the pattern cards alone, find a square of four for each of the nine groups the cards can make: **4*2**, **2222**, **2*22**, **22***, ******, **××**, **22×**, **442** and **o**. Check against the target backs.

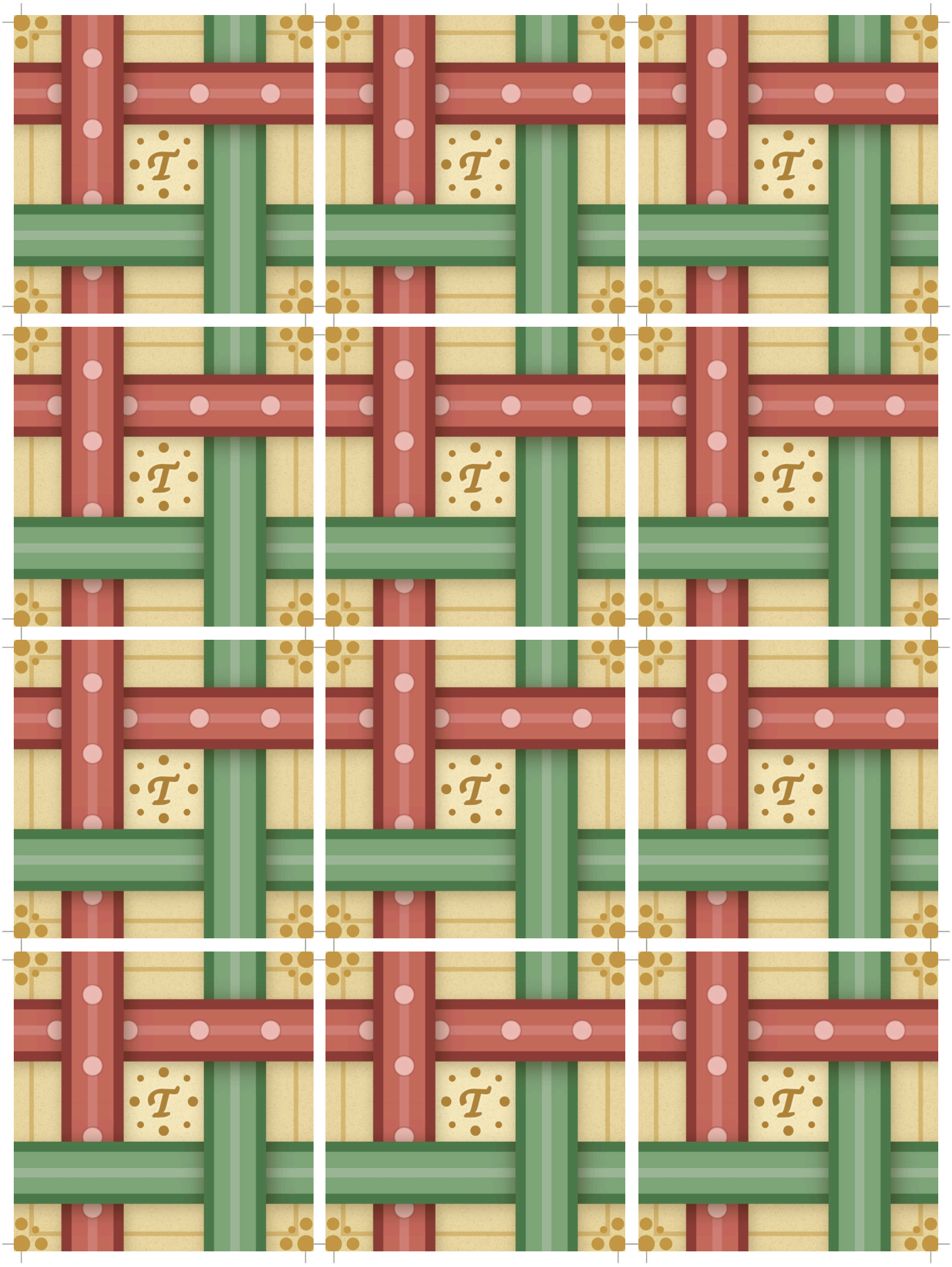
The target backs are the answers. They are printed separately, as the solutions, so the cards can be played without them; print them on the back of sheet 5 to make double-sided targets.

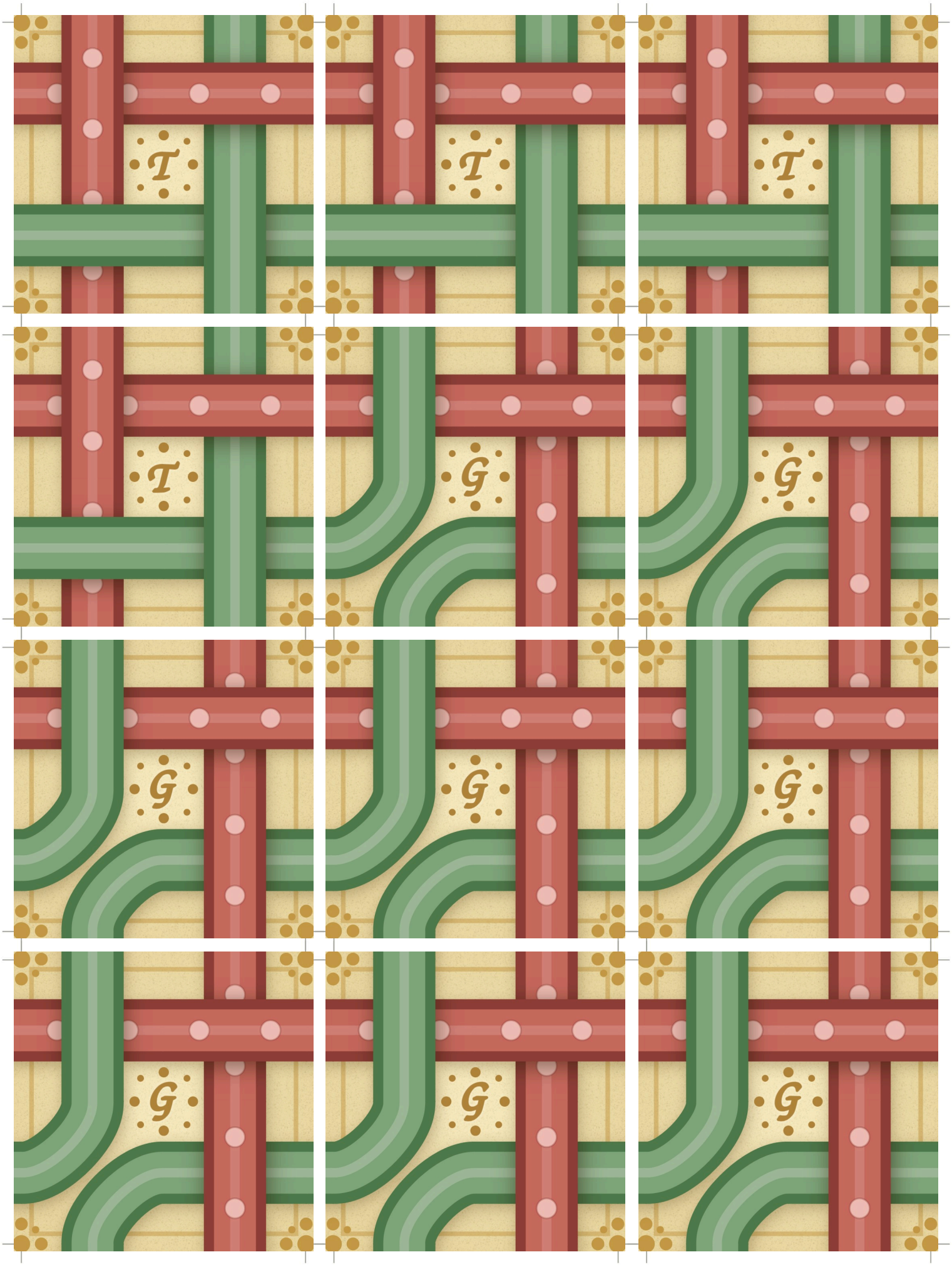
READING THE NAMES

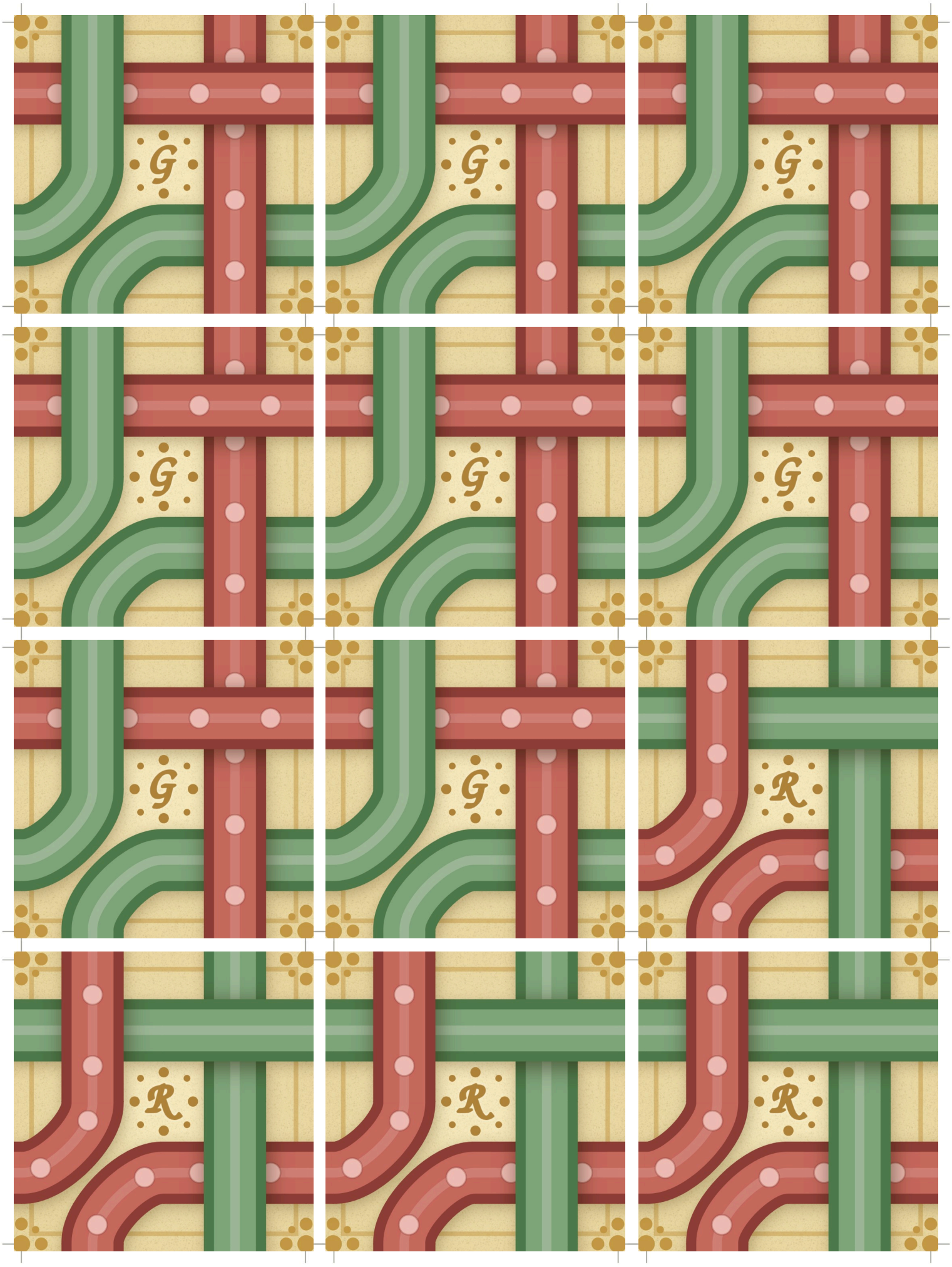
2, 4 before *	a kind of turn centre on no mirror: 2 a half turn, 4 a quarter turn
*****	mirrors: one kind, or several kinds that meet
2 after *	a kind of point where mirrors cross; the digit counts the mirrors
×	a glide (reflect, then slide) that crosses no mirror
o	nothing but repeats

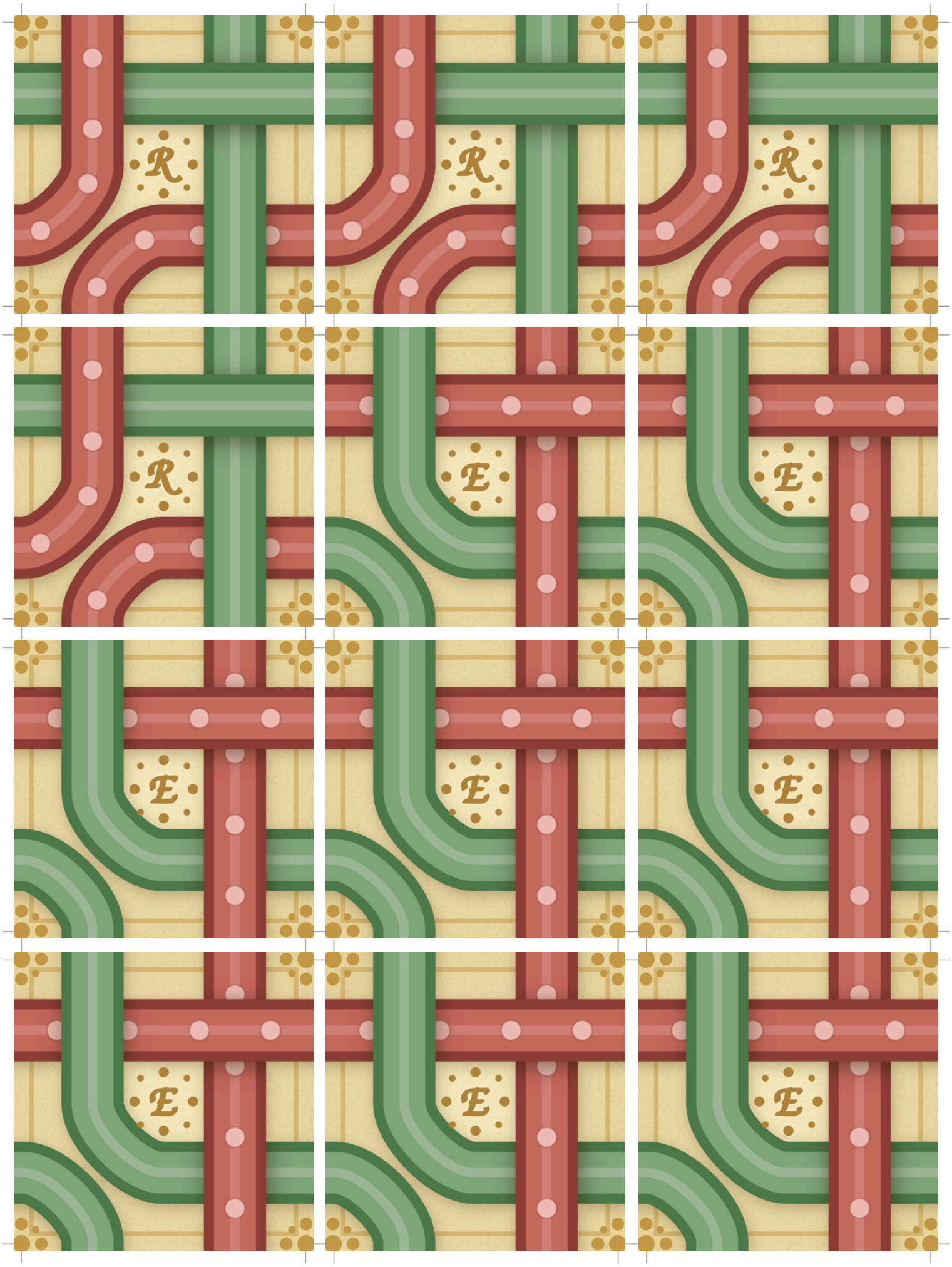
Count kinds, not copies: two marks are one kind when a symmetry of the pattern carries one onto the other. A guide to every target, step by step, and a card table to try layouts on screen: **knotworthy.co.uk/K2026-06**.

Free to print from knotworthy.co.uk/K2026-06 · CC BY 4.0




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Repeating Knot Cards

four cards · nine wallpaper groups

1. Pick a target card, recipe side up.
2. Lay out its four cards in a square, each turned so its letter faces as shown.
3. Make four such squares and join them: sixteen cards. Match ribbon colours across every edge.
4. Find the mirrors, glides and turns. Which group is it?
5. Turn the target over to check.

READING THE BACKS

- mirror
- glide: reflect and slide
- half-turn centre
- quarter-turn centre

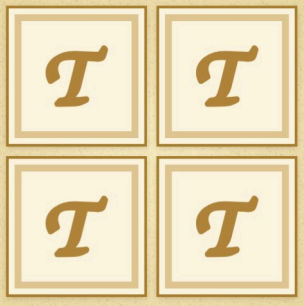
The name lists what a pattern has: a number for each kind of turn centre; ***** for mirrors, with a number after it for each kind of corner where mirrors meet; **×** for a glide that crosses no mirror; **o** for repeats only. In brackets: the crystallographers' name.

TO WONDER ABOUT

1. Why can't these cards make ***×**, the tenth group a square grid allows?
2. Change one card at a time. Can you visit all nine groups without repeating one?
3. How many strands of each colour does each pattern have?

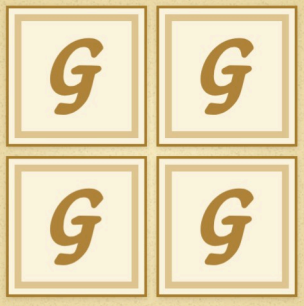


TARGET 1



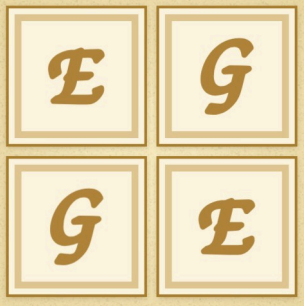
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 2



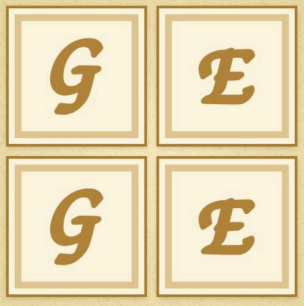
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 3



Make this square, then four of them. Which group is it? Turn over to check.

TARGET 4



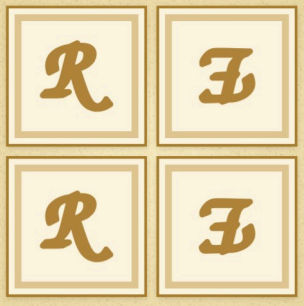
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 5



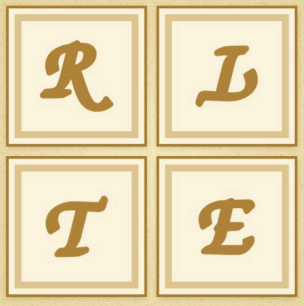
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 6



Make this square, then four of them. Which group is it? Turn over to check.

TARGET 7



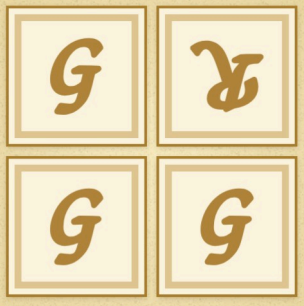
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 8



Make this square, then four of them. Which group is it? Turn over to check.

TARGET 9



Make this square, then four of them. Which group is it? Turn over to check.

THE BOX

To make it: glue this page onto thin card, such as a cereal box. Cut round each net on the solid line. Score and fold every dashed line, the coloured side out. Glue each flap inside the wall it meets. The lid fits over the tray. Sized for the 60 cards printed on 300 gsm card.

LID



TRAY

